Khulna University of Engineering & Technology, Khulna
Department of Electronics and Communication Engineering

Project Proposal

Design and Implementation of a 3-Digit Random Number Generator

Submitted By
Md. Baized Al Nur
roll: 2309033
Debroop Ghosh Dostider
roll: 2309034

Course Code: ECE 2104

Course Title: Digital Electronics and Logic Circuits Laboratory OPEN ENDED LAB

CamScanner

Introduction

Random number generation is a fundamental requirement in many digital applications, ranging from lottery systems and gaming devices to security and industrial automation. While software-based random number generators are commonly implemented using microcontrollers or processors, it is equally possible — and educationally valuable — to realise such a system entirely through discrete digital ICs. This project presents the design and implementation of a **Digital Lottery Machine**, a three-digit Random Number Generator (RNG) built without any microcontroller. The system leverages the NE555 Timer IC configured as an astable oscillator and three cascaded CD4026 decade counter/display driver ICs, each driving a common cathode 7-segment display. When the user presses a push-button, the high-frequency clock increments the counters rapidly; upon release, the count freezes instantly, producing a pseudo-random three-digit number (000–999). This design demonstrates core concepts of digital electronics — oscillators, counters, and display drivers — in a compact, cost-effective, and microcontroller-free implementation.

Objectives

1. To design and implement a three-digit random number generator circuit using only discrete digital ICs — specifically the NE555 Timer and CD4026 decade counter/driver — without any microcontroller.
2. To configure the NE555 Timer in astable mode to generate a stable, high-frequency clock pulse train for driving the counter stages.
3. To cascade three CD4026 ICs to count independently from 0 to 9 on each digit, collectively producing a three-digit output (000–999) on common cathode 7-segment displays.
4. To implement a push-button controlled clock-gating mechanism that allows the user to freeze the counter at any instant, simulating a random number selection.
5. To simulate the complete circuit in Proteus Design Suite prior to physical assembly, verifying correct oscillation, counting, carry propagation, and latch behaviour.
6. To demonstrate the practical application of fundamental digital electronics components in building a real-world, functional lottery machine.

CamScanner

Problem Statement

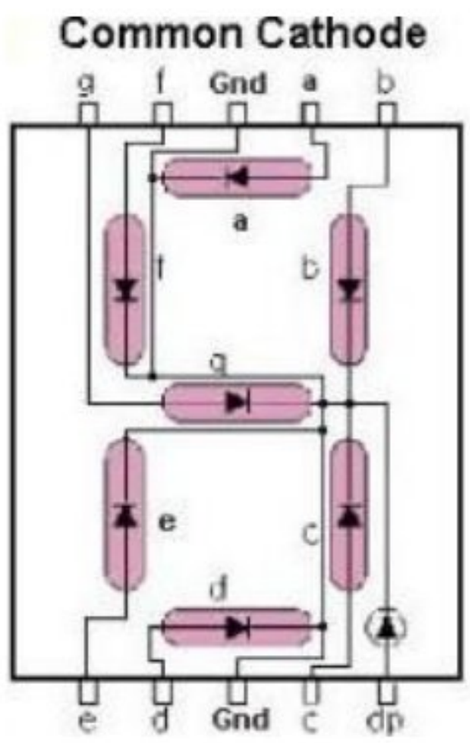
Design a three-digit random number generation system without any microcontroller. When a momentary push-button is pressed and released, the system must latch a pseudo-random three-digit number (000–999) and display each digit independently on a dedicated common cathode 7-segment display. The design must rely solely on the **NE555 Timer** (astable oscillator) and **CD4026** (decade counter + 7-segment driver) ICs for counting and display driving — no microcontroller, FPGA, or additional decoder ICs are permitted. Each digit must be selected randomly and displayed independently. The generated number must lock in until the next trigger.

Main Components

Main components employed in the lab solution are as under:

7-Segment Display

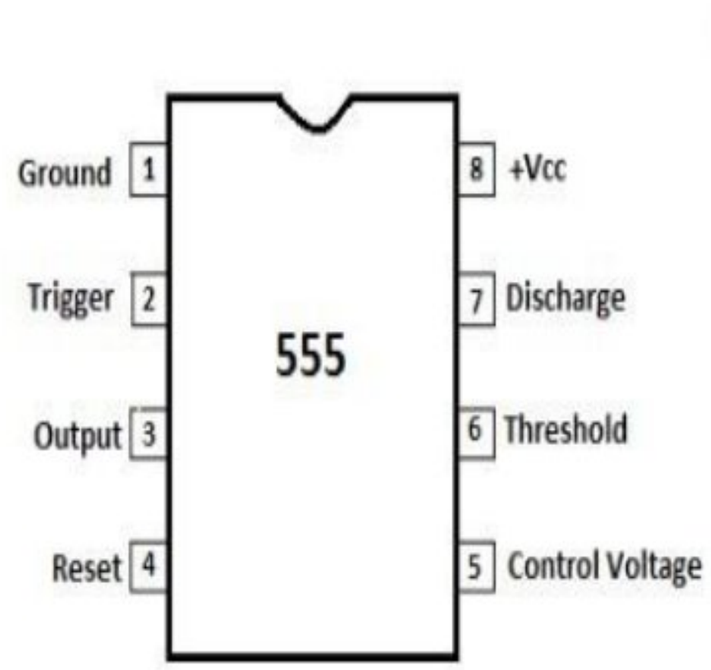
A seven-segment display is a form of electronic display device for displaying decimal numerals that is an alternative to the more complex dot matrix displays. **Seven-segment displays are** widely used in digital clocks, electronic meters, basic calculators, and other electronic devices that display numerical information. Common cathode variants share a single ground connection and are directly compatible with the CD4026 output.



CamScanner

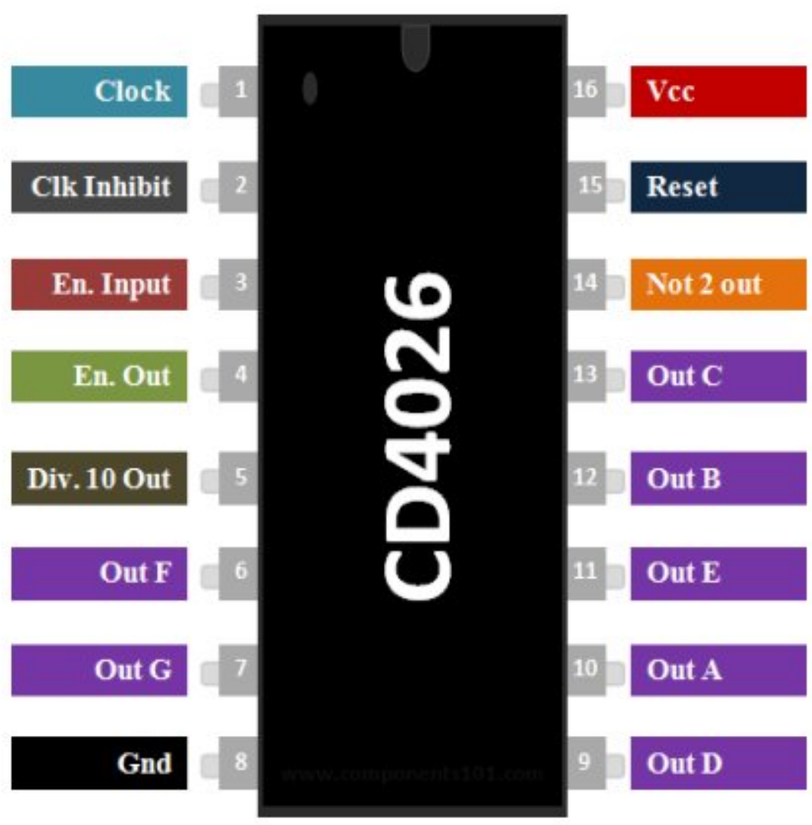
NE555 Timer

The 555 timer IC is an integrated circuit used in a variety of timer, pulse generation, and oscillator applications. The 555 can be used to provide time delays, as an oscillator, and as a flip-flop element. In this design it is configured in **astable mode**, generating a continuous high-frequency clock pulse train whose frequency is set by two 10 kΩ resistors and a 10 nF capacitor.



CD4026 IC

The **IC CD4026** is an IC which can perform the function of both a counter as well as a **7-segment Driver**. One single IC can be used to count from zero (0) to nine (9) directly on a Common Cathode type 7-segment display. The count can be increased by simply giving a high clock pulse; also more than one digit (0–9) can be created by cascading more than one **CD4026 IC**. Three ICs are cascaded — Units, Tens, and Hundreds — using the Carry Out pin of each stage to clock the next.



CamScanner

Push-Button Switch (×1)

A momentary normally-open push-button acts as the user trigger. Pressing it enables the NE555 clock to reach the CD4026 counters (counters run freely); releasing it disconnects the clock, freezing all three displays on a pseudo-random three-digit number.

Resistors — 10 kΩ (×2) and 1 kΩ (×1)

The two 10 kΩ resistors set the charge/discharge timing of the NE555 astable oscillator, determining the clock frequency. The 1 kΩ resistor serves as a current-limiting element in the output/enable path.

Capacitors —0.01 µF

The 0.01 uF capacitor is the main timing capacitor of the NE555 oscillator. The 20 pF capacitor is a bypass/noise-decoupling capacitor placed on the control voltage pin (pin 5) of the NE555 to suppress high-frequency interference and stabilise oscillation.

Table 1: Summary of Components Used

Component	Part No. / Value	Qty	Function in Circuit
NE555 Timer IC	NE555	1	Astable oscillator – generates clock pulses
Decade Counter / 7-Seg Driver	CD4026	3	Counts 0–9 and directly drives each display digit
Common Cathode 7-Seg Display	CC type	3	Displays Units, Tens, and Hundreds digits
Push-Button Switch	SPST NO	1	User trigger – enables/freezes the clock
Resistor	10 kΩ,1 kΩ,	2	Sets NE555 timing frequency (R1, R2)
Resistor	1 kΩ	1	Current limiting in output path
Capacitor	0.01 µF (103)	1	Main timing capacitor of NE555 oscillator
Capacitor	0.01µF (200)	1	Noise decoupling on NE555 control pin (pin 5)

CamScanner

Working Principle / Circuit Operation

The circuit exploits the speed of digital electronics to achieve randomness. The following steps describe the complete operation of the system:

- When the **push-button is pressed**, the NE555 Timer configured in astable mode begins generating a continuous square-wave clock pulse train. The frequency of oscillation is determined by the values of R1 (10 kΩ), R2 (10 kΩ), and C1 (10 nF), producing a clock rate in the kilohertz range — far beyond human perception.
- The clock pulses are gated through the push-button connection and fed directly into the **CLK IN (Pin 1)** of the first CD4026 IC, which represents the **Units digit**. This IC increments its internal decade counter by one on each rising clock edge, updating the units digit on Display 1 at high speed.
- When the first CD4026 completes a full count cycle (0 through 9), it produces a pulse on its **Carry Out (Pin 5)**. This carry pulse is fed directly into the CLK IN of the second CD4026 IC, which represents the **Tens digit**. The tens display increments accordingly.
- Similarly, the Carry Out of the second CD4026 clocks the third CD4026 IC, representing the **Hundreds digit**. This cascaded arrangement allows the three-stage counter to count from 000 to 999 in a continuous rolling sequence for as long as the push-button is held.
- When the **push-button is released**, the clock signal path is immediately interrupted. Each CD4026 IC retains its last internal count, and all three 7-segment displays freeze on the digit they last showed — producing a locked three-digit number.
- Because the clock operates at a frequency several orders of magnitude faster than human reaction time, the exact moment of release — and therefore the frozen number — is effectively **unpredictable to the user**, simulating a genuine random number generation.
- The **20 pF decoupling capacitor** on pin 5 of the NE555 ensures stable oscillation by filtering any noise on the control voltage line, preventing false triggering or irregular pulse widths during operation.

CamScanner

Proteus Implementation

The complete circuit will be designed and simulated in **Proteus Design Suite** prior to hardware assembly. The schematic will incorporate all components with their exact values, labelled nets, and power rails. Virtual oscilloscope probes will be placed on the NE555 output (Pin 3) and the Carry Out pins of each CD4026 stage to capture and verify waveforms. The following key aspects will be validated through simulation:

- The NE555 astable oscillator will be confirmed to produce a stable square-wave output at the expected frequency.
- Correct decade counting behaviour and carry-chain propagation across all three cascaded CD4026 stages will be verified.
- Accurate segment-output mapping to all three 7-segment displays for digits 0–9 will be checked.
- The latch behaviour — counter freezing instantly upon push-button release — will be demonstrated through repeated simulation runs.

Advantages of the Proposed System

proposed microcontroller-free random number generator offers several notable advantages over software-based or microcontroller-driven alternatives:

- **No Microcontroller Required:** The entire system operates using standard CMOS and bipolar ICs, eliminating the need for programming, firmware, or a development environment. This reduces cost and complexity significantly.
- **Low Cost and Wide Availability:** The NE555 and CD4026 are among the most widely available and inexpensive ICs in the market, making the circuit highly accessible for educational and low-budget applications.
- **Simple and Robust Design:** With minimal components and no software layer, the circuit has very few failure points. Its straightforward topology makes troubleshooting and maintenance easy.
- **True Real-Time Response:** The clock-gating mechanism responds instantaneously to the push-button, with no processing delay or interrupt latency — unlike microcontroller-based systems.

CamScanner

- **Educational Value:** The design provides an excellent practical demonstration of astable oscillators, decade counters, carry-chain cascading, and 7-segment display driving — core topics in any digital electronics curriculum.
- **Scalable Design:** Additional CD4026 stages can be cascaded effortlessly to extend the random number range beyond three digits without any circuit redesign.
- **Low Power Consumption:** CMOS-based CD4026 ICs consume very little standby current, making the system suitable for battery-powered applications.

References

[1] Texas Instruments. (2015). *NE555 Precision Timer Datasheet*. Document No. SNAS548D.

[2] Texas Instruments. (1998). *CD4026B CMOS Decade Counter with 7-Segment Display Outputs Datasheet*.

[3] Mano, M. M., & Ciletti, M. D. (2013). *Digital Design* (5th ed.). Pearson Education.

[4] Floyd, T. L. (2015). *Digital Fundamentals* (11th ed.). Pearson.

Fairchild Semiconductor. (2000). *LM555/LM555C Timer Datasheet*.

CamScanner